

Finding the Bus's Distance from its Route

The 'fast formula' explained, with worked numbers

The route is a line on the map; the bus is somewhere off it. We want the perpendicular distance — the shortest gap between the bus and the road.

Step 1 — Pretend the Earth is flat

Over a few kilometres, Earth's curvature is invisible. So we convert each latitude/longitude into plain (x, y) metres, measured from a reference point near the route:

$$x = R \cdot (\lambda - \lambda_0) \cdot \cos \varphi_0 \quad (\text{metres east of reference})$$

$$y = R \cdot (\varphi - \varphi_0) \quad (\text{metres north of reference})$$

R = Earth's radius, 6,371,000 m. The $\cos \varphi_0$ corrects for the fact that a degree of longitude is shorter away from the equator. After this conversion, every point is just (x, y) in metres and we can use regular geometry.

Step 2 — Example

Reference point = Stop P_0 (so P_0 itself sits at the origin).

Point	x (m)	y (m)	What it means
P_0	0	0	The reference itself
P_1	-1192	+1133	1192 m west, 1133 m north of P_0
Bus 17	-613	+556	613 m west, 556 m north of P_0

Distance from P_0 to P_1 : $\sqrt{1192^2 + 1133^2} \approx 1645$ m.

Step 3 — Find the closest point on the segment to the bus

From P_0 , draw two arrows:

- **Arrow S** = P_0 to P_1 (along the road) = $(-1192, +1133)$
- **Arrow V** = P_0 to bus = $(-613, +556)$

The fraction of the way along the segment that gets you closest to the bus is:

$$t^* = \frac{\vec{V} \cdot \vec{S}}{\|\vec{S}\|^2}$$

where $\vec{V} \cdot \vec{S}$ is the dot product (multiply matching components, then add) and $\|\vec{S}\|^2$ is the segment length squared. **In plain English: how much of arrow V points the same direction as arrow S.**

Numbers:

- $\vec{V} \cdot \vec{S} = (-613)(-1192) + (556)(1133) = 730\,696 + 629\,948 = 1\,360\,644$
- $\|\vec{S}\|^2 = 1192^2 + 1133^2 = 2\,704\,553$
- $t^* = 1\,360\,644 / 2\,704\,553 \approx \mathbf{0.503}$

So the bus is **50.3% of the way along the segment** — basically at the midpoint.

The `clip(..., 0, 1)` step in the original formula just snaps to an endpoint if the bus is past either end of the segment. With $t^* = 0.503$ we're already inside $[0, 1]$, so no clipping happens.

Step 4 — Where is that closest point, and how far is the bus from it?

$$\text{closest point} = P_0 + 0.503 \times (P_1 - P_0) = (-600, +570)$$

$$\text{offset} = (-613 - (-600), 556 - 570) = (-13, -14)$$

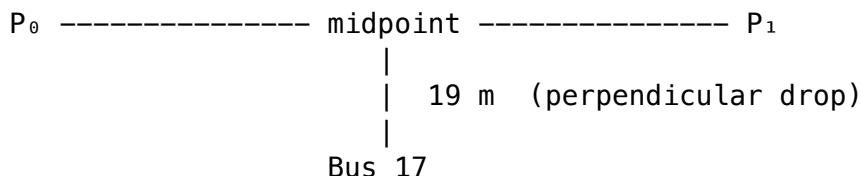
$$\delta = \sqrt{13^2 + 14^2} = \sqrt{365} \approx \mathbf{19\,m}$$

Step 5 — Verdict

Bus 17 is **19 m off the centre line** of its route. Our corridor is 50 m wide.

$19 \leq 50$, so the bus is on route. Carry on.

The picture



The dot-product trick slides the bus's position sideways onto the road, then measures the leftover gap. That gap is δ . If $\delta \leq 50$ m the bus is on route; if it exceeds 50 m for **three consecutive packets**, we raise an OFF_ROUTE alert (and the divert-alert pipeline fires).